

Sign-Balanced Liquidation Clearing in DeFi Markets

Abstract. Liquidations in DeFi lending protocols are interdependent: selling collateral moves prices, which can trigger or suppress other liquidations. We study a collateral-only liquidation model with linear price impact, in which liquidators repay debt externally and only collateral sales move market prices. Even in this conservative mechanism, clearing can be nonmonotone: when an asset sold as collateral is also used as another position’s debt asset, its price decline reduces that borrower’s debt burden.

We show that sign-balance has a precise complexity-theoretic meaning. If the asset borrowing graph is bipartite, then after flipping one side of the asset prices and complementing liquidation variables on one side, the smooth clearing equations become a subtraction-free monotone clipped-affine system. Approximate clearing in this sign-balanced regime reduces to the succinct Tarski fixed-point problem and therefore lies in $\text{CLS} = \text{PPAD} \cap \text{PLS}$. In the small-impact regime, the same system is a contraction and clearing is computable in polynomial time. Conversely, once sign-balance is removed, non-bipartite instances can implement a NOT primitive, encode SUM/NOT generalized circuits, and finding even a constant-approximate clearing is PPAD-hard. Smooth approximate clearing is in PPAD, and all-or-nothing clearing is in NP with a polynomial-time balanced case.

Keywords: DeFi liquidation · Clearing · PPAD · CLS · Tarski fixed points · Financial networks

1 Introduction and Technical Overview

DeFi lending protocols liquidate undercollateralized positions. A position has a collateral asset and a debt asset. If it becomes unhealthy, liquidators repay debt and receive collateral; at the market level, the relevant price impact is that collateral is sold. When positions are large relative to liquidity, these sales move prices, and those price movements affect the health of other positions. A clearing is a self-consistent vector of liquidations and prices: the liquidations implied by prices must be exactly the liquidations whose collateral sales generate those prices.

This paper studies the computational problem created by this feedback. We use a linear price-impact market as a first-order model of liquidation-induced price movement. This modeling choice is deliberately conservative. Liquidations only sell collateral; debts are repaid externally. There is no market buy pressure on the debt asset. Thus the nonmonotonicity we identify does not arise from an artificial assumption that liquidators buy debt in the same market. It arises from the fact that the same asset can appear as collateral in one position and as debt in another.

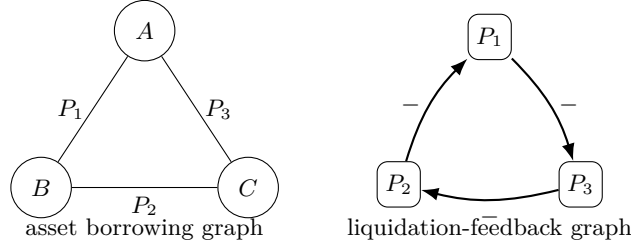


Fig. 1. A three-asset collateral-debt cycle. Selling an asset can lower another position’s debt value, creating a negative liquidation dependency.

A DeFi-native negative feedback. Consider three assets A, B, C and three positions

$$P_1 = (A, B), \quad P_2 = (B, C), \quad P_3 = (C, A),$$

where (A, B) denotes collateral A and debt B . If P_1 is liquidated more aggressively, more A is sold and p_A falls. But A is the debt asset of P_3 , so the value of P_3 ’s debt falls. Thus P_3 becomes healthier and may liquidate less. A liquidation can therefore rescue another borrower whose debt is denominated in the asset being sold. This effect creates an all-negative cycle of liquidation feedback; see Fig. 1. It is the primitive behind both our sign-balance theorem and our hardness gadgets.

Positioning relative to monotone clearing. Financial clearing problems often have monotone or lattice structure. Eisenberg–Noe type payment networks are the classical example [14], and recent work makes the Tarski connection explicit: Besting, Hoefler, and Huth compute extremal Tarski fixed points in generalized financial networks with monotone piecewise-linear payment functions and default costs [1]. Fire-sale games also exhibit lattice equilibrium structure under convex price impact [2]. Our contribution is therefore not the observation that clearing can be phrased as a Tarski fixed point. The contribution is to identify what changes in DeFi liquidation clearing. The map here is not a conservation-based payment allocation; it is a price-impact fixed point in which liquidations change asset prices, and an asset sold as collateral can simultaneously be another borrower’s debt asset. This debt-relief channel supplies a negative feedback primitive. We show that the collateral-debt graph determines whether this primitive can be globally aligned into a monotone Tarski/CLS problem or can instead implement SUM/NOT circuits and PPAD-hard fixed-point structure.

Main results. Let G_A be the asset borrowing graph: its vertices are assets and each position gives an edge between its collateral and debt assets. Our first main result identifies when the DeFi liquidation map preserves the Tarski-type structure familiar from monotone clearing.

Sign-balanced side. If G_A is bipartite, smooth δ -clearing reduces to the succinct Tarski fixed-point problem and belongs to $\text{CLS} = \text{PPAD} \cap \text{PLS}$.

If, in addition, liquidation feedback is small enough to make the transformed map contractive, clearing is unique and computable in polynomial time.

The proof uses the same sign transform as a monotonicity argument, but we do not claim that monotonicity alone gives a polynomial-time algorithm. After flipping one side of the asset prices and complementing liquidation variables whose collateral lies on one side, the equations become

$$y = \Pi_{[0,u]}(Ay + c), \quad A \geq 0.$$

Thus the transformed map is order-preserving. Discretizing it on a grid yields a succinct monotone Tarski instance; a grid fixed point is an approximate clearing. The same representation also shows that the balanced equations have an upper-bounded Z-matrix complementarity structure. This structural reduction is useful, but by itself it is not a polynomial-time algorithm in the large-impact regime.

Our negative result gives the other side of the complexity map.

Non-bipartite worst case. There exists a constant $\delta_0 > 0$ such that, in the non-bipartite regime, finding a δ_0 -approximate clearing is PPAD-hard.

We reduce from generalized circuits with only truncated-SUM and NOT gates, a restricted source problem known to be PPAD-complete for a constant approximation parameter. The reduction first ensures that the source circuit has an intrinsic odd-NOT cycle in the same dependency component as a hard-instance wire. The liquidation gadgets preserve this parity: NOT gates become odd collateral-debt paths, while SUM gates become even paths. Thus the hard instances are non-bipartite for the same reason that the source circuits are sign-frustrated. Finally, smooth approximate clearing belongs to PPAD because the map is rational piecewise-linear, all-or-nothing clearing existence belongs to NP, and the balanced all-or-nothing case is polynomial-time solvable by finite monotone iteration.

The paper’s contributions can be summarized as follows.

- *DeFi-native model*: a collateral-only liquidation clearing problem with linear price impact, close-factor-compatible sales, and debt relief through asset-denominated liabilities.
- *Sign-balance criterion*: bipartite asset borrowing graphs yield subtraction-free monotone clipped-affine fixed points, hence CLS membership for smooth approximate clearing.
- *Two-dimensional complexity map*: small feedback gain gives contraction and polynomial-time computation, while large-impact bipartite clearing remains open.
- *Hardness primitive*: odd-NOT frustration in SUM/NOT circuits maps to non-bipartite collateral-debt cycles, giving constant- δ PPAD-hardness in the large-impact worst case.

Table 1. Complexity map. The hard entries are worst-case statements; small-impact entries are sufficient conditions for polynomial-time convergence.

Sign-balanced / bipartite	Non-bipartite
Small impact unique clearing in P	unique clearing in P under contraction
Large impact in CLS via Tarski; P open	constant- δ PPAD-hard in worst case

Relation to prior work. Prior work establishes Tarski/lattice structure, and in some settings polynomial algorithms, for conservation-like payment networks and strategic fire-sale models [1, 2]. We use these models as a foil. Our question is which DeFi network structures preserve Tarski-type monotonicity and which structures destroy it strongly enough to encode PPAD-hard fixed points. Credit-default-swap clearing hardness is another important antecedent [16, 4], but its nonmonotonicity comes from default-contingent derivative payments. Here, the NOT primitive comes from collateral-only liquidation lowering the price of another position’s debt asset.

What the dichotomy does and does not say. The hardness theorem is a worst-case statement outside the balanced class. It does not say that every non-bipartite instance is hard. If price impact is sufficiently small, the clearing map may be a contraction even on a non-bipartite graph. The correct interpretation is structural: bipartiteness gives an order-preserving Tarski structure and CLS membership, while non-bipartite networks can encode PPAD-hard fixed-point structure when price impact is large enough.

2 Model and Sign Structure

There are assets $k \in [n]$, prices $p_k \in [0, M_k]$, and positions $i \in [m]$. Position i has collateral asset a_i , debt asset b_i , collateral amount $C_i > 0$, debt amount $D_i > 0$, liquidation threshold $\theta_i \geq 1$, and smoothing width $\varepsilon_i > 0$. We assume $a_i \neq b_i$. The health deficit is

$$h_i(p) = \theta_i D_i p_{b_i} - C_i p_{a_i}. \quad (1)$$

A positive deficit means the position is unhealthy. In the smooth model, liquidation intensity is

$$x_i = \Pi_{[0,1]} \left(\frac{h_i(p)}{\varepsilon_i} \right). \quad (2)$$

Liquidation sells collateral only. Thus the price equation is

$$p_k = \Pi_{[0,M_k]} \left(p_k^0 - s_k \sum_{i:a_i=k} \alpha_i x_i \right), \quad (3)$$

where $s_k > 0$ is the impact slope and $\alpha_i \geq 0$ is the collateral amount sold by a fully liquidated position. When we impose a close-factor constraint, we require

$0 \leq \alpha_i \leq \varphi C_i$ for a fixed $\varphi \in (0, 1]$. Debt is repaid externally and does not create a market buy order. The clipping makes the clearing map a continuous self-map of a compact box. The hardness construction below respects the close-factor bound; it adjusts output weights through market-depth parameters s_k , not by selling more collateral than a position contains.

A smooth exact clearing is a pair (p, x) satisfying the price and liquidation equations. A δ -clearing satisfies

$$\max\{\|p - P(x)\|_\infty, \|x - X(p)\|_\infty\} \leq \delta,$$

where P and X denote the right-hand sides of (3) and the liquidation equation. In the all-or-nothing model, $x_i \in \{0, 1\}$; a clearing is a threshold-consistent liquidation set.

All parameters in the input are rational numbers encoded in binary. The computational problems ask for a δ -approximate clearing for a rational approximation parameter δ ; in the contraction regimes below, exact or high-precision approximate clearings can be obtained by iteration. In the hardness theorem, δ is a fixed positive constant determined by the source generalized-circuit hardness constant and by the constant error amplification of the gadgets. The price clipping bounds M_k are part of the input; in the reduction they are constants.

We use two graphs. The asset borrowing graph G_A has assets as vertices and an undirected edge $\{a_i, b_i\}$ for each position. This is the graph used in the tractability and hardness statements. The position signed graph G_X has positions as vertices; it has a positive directed edge $j \rightarrow i$ when $a_j = a_i$, and a negative directed edge $j \rightarrow i$ when $a_j = b_i$. The second graph is diagnostic: it explains local liquidation feedback and the gadgets.

Economic meaning of bipartiteness. Bipartiteness should not be read as an arbitrary coloring condition. It says that assets can be separated into two sectors and all borrowing positions cross the sector boundary. A motivating special case is a market in which one sector contains volatile collateral-like assets and the other contains stable or debt-like assets, with positions of the form “deposit volatile, borrow stable” and “deposit stable, borrow volatile.” The theorem does not require this interpretation, but it explains why bipartiteness is economically meaningful: it rules out odd cycles in which an asset returns with the opposite collateral/debt role, the source of the debt-relief negative feedback in Fig. 1.

Lemma 1 (price-level sign). *In the collateral-only model, the price map is competitive on the asset borrowing graph: if $m \neq k$, increasing p_m can only weakly decrease the next price of k , and only through positions with collateral k and debt m .*

Proof. For a position with collateral k and debt m , increasing p_m increases h_i , hence weakly increases x_i . This increases sales of asset k and weakly decreases the next price of k . Positions not using m as debt do not create this cross-effect. The clipping maps are nondecreasing, so the conclusion holds without differentiability assumptions.

At the liquidation-intensity level, if $a_j = a_i$, more liquidation of j depresses p_{a_i} , making i less healthy and increasing x_i : a positive dependency. If $a_j = b_i$, more liquidation of j depresses p_{b_i} , making i 's debt cheaper and decreasing x_i : a negative dependency. Thus odd cycles of such negative dependencies are the obstruction to sign-monotonicity. The three-asset example in Fig. 1 is an all-negative directed 3-cycle in G_X .

This diagnostic graph is useful for intuition, but the algorithmic boundary is stated on the asset graph G_A . In generic nondegenerate regions, a signed coordinate flip can make all direct liquidation dependencies monotone exactly when the signed feedback graph has no odd negative cycle, in the sense of Harary balance. The asset-level bipartite condition is the clean sufficient condition we use throughout the paper: every collateral-debt edge crosses the cut, so one global sign convention makes the competitive price interactions coherent.

If G_A is bipartite, a coordinated sign flip removes the competitive signs at the price level. More explicitly, the bipartition defines an orthant order on prices: prices on one side are ordered normally and prices on the other side are ordered in reverse. In that order the price map is isotone. A parallel transform on liquidation variables gives the sign-balanced order used below.

Sign balance versus hardness. The two graph views should not be conflated. The asset graph G_A controls the sign pattern of the price map, and bipartiteness of G_A is the structural assumption used on the sign-balanced side. The position graph G_X records local liquidation-to-liquidation feedback: co-writers with a common collateral asset produce positive edges, while a writer whose collateral is another position's debt asset produces a negative edge. Harary's balance criterion says that a signed graph admits a coordinate flip making all edges positive if and only if it has no cycle with an odd number of negative edges. The triangle in Fig. 1 violates this criterion.

The positive and negative results therefore have different quantifiers. Bipartiteness gives a monotone Tarski structure and places approximate clearing in CLS; it gives a polynomial-time algorithm under the additional small-impact condition of Proposition 2. Non-bipartiteness is a worst-case source of hardness: outside the balanced class, large-impact instances can be constructed that encode PPAD-complete fixed-point structure. Small-impact non-bipartite instances may still be contractions and hence easy. Sign balance is therefore a complexity-class certificate rather than an instance-level "easy if and only if" criterion.

The distinction between monotone structure and polynomial-time tractability is important. Tarski's theorem implies existence of fixed points and, in the succinct setting, a well-defined total search problem, but it does not by itself give a polynomial-time algorithm for a continuous map. The next section uses the monotone structure to place sign-balanced clearing in CLS, and then identifies a separate small-impact condition under which ordinary contraction gives polynomial-time computation.

3 Sign-Balanced Clearing, Tarski, and Small-Impact Tractability

Theorem 1. *In the collateral-only linear price-impact model with price clipping, if the asset borrowing graph G_A is bipartite, then for every rational $\delta > 0$, smooth δ -clearing reduces in polynomial time to the succinct Tarski fixed-point problem. Hence bipartite smooth δ -clearing belongs to $\text{CLS} = \text{PPAD} \cap \text{PLS}$.*

Let (L, R) be a bipartition of G_A . Define transformed price variables

$$q_k = \begin{cases} p_k, & k \in L, \\ M_k - p_k, & k \in R, \end{cases} \quad (4)$$

and transformed liquidation variables

$$z_i = \begin{cases} 1 - x_i, & a_i \in L, \\ x_i, & a_i \in R. \end{cases} \quad (5)$$

Let $y = (q, z)$ and let $[0, u]$ be the corresponding box, with upper bounds M_k for transformed prices and 1 for transformed liquidations.

Lemma 2. *Under the above transform, the clearing equations are equivalent to*

$$y = \Pi_{[0, u]}(Ay + c) \quad (6)$$

for a rational matrix $A \geq 0$ and vector c . Moreover, after ordering price and liquidation variables separately, A has block form

$$A = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix}, \quad P, Q \geq 0.$$

Proof. There are four cases. First, if $k \in L$, then every position with collateral k has $x_i = 1 - z_i$, and

$$q_k = p_k = \Pi_{[0, M_k]} \left(p_k^0 - \sum_{i: a_i = k} s_k \alpha_i + \sum_{i: a_i = k} s_k \alpha_i z_i \right),$$

which has nonnegative coefficients on the z_i 's. Second, if $k \in R$, then $q_k = M_k - p_k$ and $x_i = z_i$ for positions collateralized by k . Using $M - \Pi_{[0, M]}(r) = \Pi_{[0, M]}(M - r)$,

$$q_k = \Pi_{[0, M_k]} \left(M_k - p_k^0 + \sum_{i: a_i = k} s_k \alpha_i z_i \right),$$

again with nonnegative coefficients.

For liquidations, consider $a_i \in L, b_i \in R$. Then $p_{a_i} = q_{a_i}$, $p_{b_i} = M_{b_i} - q_{b_i}$, and $z_i = 1 - x_i$. Since $1 - \Pi_{[0,1]}(r) = \Pi_{[0,1]}(1 - r)$,

$$z_i = \Pi_{[0,1]} \left(1 - \frac{\theta_i D_i M_{b_i}}{\varepsilon_i} + \frac{\theta_i D_i}{\varepsilon_i} q_{b_i} + \frac{C_i}{\varepsilon_i} q_{a_i} \right).$$

Finally, if $a_i \in R, b_i \in L$, then $z_i = x_i$, $p_{a_i} = M_{a_i} - q_{a_i}$, and $p_{b_i} = q_{b_i}$, giving

$$z_i = \Pi_{[0,1]} \left(-\frac{C_i M_{a_i}}{\varepsilon_i} + \frac{\theta_i D_i}{\varepsilon_i} q_{b_i} + \frac{C_i}{\varepsilon_i} q_{a_i} \right).$$

All displayed coefficients of variables are nonnegative. Price equations depend only on transformed liquidation variables, and liquidation equations depend only on transformed price variables, which gives the block-off-diagonal form.

The map

$$T(y) = \Pi_{[0,u]}(Ay + c)$$

is therefore isotone in the product order. This is the true algorithmic consequence of sign-balance. It is stronger than a qualitative sign observation but weaker than a polynomial-time algorithm in the large-impact continuous setting: Tarski guarantees fixed points of monotone maps, and succinct monotone fixed-point search has its own complexity class behavior.

Proof (Proof of Theorem 1). Fix $\delta > 0$. Choose a rational grid step $\eta \leq \delta$ with polynomially many bits, and let

$$\mathcal{G}_\eta = [0, u] \cap \eta \mathbb{Z}^d$$

with the natural product order. Define a grid map

$$T_\eta(y) = \lfloor T(y) \rfloor_\eta,$$

where the floor is taken coordinate-wise to the largest grid point not exceeding $T(y)$. Since T is isotone and coordinate-wise rounding down is isotone, T_η is an order-preserving self-map of the finite grid lattice. It is also succinctly computable from the rational clearing instance.

A fixed point $y = T_\eta(y)$ is an η -approximate fixed point of T : coordinate-wise, $y \leq T(y) < y + \eta$, so $\|T(y) - y\|_\infty \leq \eta \leq \delta$. Transforming back gives a δ -clearing of the original market. Thus bipartite δ -clearing reduces to the succinct Tarski fixed-point problem. Etessami, Papadimitriou, Rubinstein, and Yannakakis show that the succinct Tarski problem lies in both PPAD and PLS [7]. Fearnley, Goldberg, Hollender, and Savani show that $\text{CLS} = \text{PPAD} \cap \text{PLS}$ [8]. Therefore bipartite δ -clearing belongs to CLS.

Proposition 1 (sign-balanced comparative statics). *Fix a grid used in the proof of Theorem 1. Suppose two sign-balanced instances have transformed grid maps T_η and S_η with $T_\eta(y) \leq S_\eta(y)$ for every grid point y . Then the least grid clearing of T_η is no larger than the least grid clearing of S_η , and the same monotone order holds for the greatest grid clearings.*

Proof. On a finite lattice, the least fixed point of an isotone map is obtained by iterating from the bottom element, and the greatest fixed point by iterating from the top. The pointwise inequality and isotonicity imply by induction that every bottom iteration under T_η is no larger than the corresponding bottom iteration under S_η . The top iteration gives the greatest fixed point comparison. In economic terms, parameter shocks that shift the transformed clearing map upward have ordered effects on the extremal sign-balanced clearings; in the original variables, the interpretation follows the same sign transform used in Theorem 1.

Proposition 2 (small-impact polynomial tractability). *In the transformed bipartite system $T(y) = \Pi_{[0,u]}(Ay + c)$, if $\rho(A) < 1$, then the clearing is unique and a δ -clearing can be computed in time polynomial in the input size and $\log(1/\delta)$.*

Proof. By Perron-Frobenius, since $A \geq 0$ and $\rho(A) < 1$, there are $v > 0$ and $\lambda < 1$ with $Av \leq \lambda v$. In the weighted sup norm $\|r\|_v = \max_i |r_i|/v_i$, the projection $\Pi_{[0,u]}$ is nonexpansive coordinate-wise and

$$\|T(y) - T(y')\|_v \leq \|A(y - y')\|_v \leq \lambda \|y - y'\|_v.$$

Thus T is a contraction. Banach iteration from any point converges geometrically to the unique fixed point; with rational data, the standard bit-complexity bound for iterating a contraction gives a δ -clearing in time polynomial in the input and $\log(1/\delta)$.

The transformed equations also have a useful complementarity interpretation. Let $B = I - A$ and $d = -c$. Then $y = T(y)$ is equivalent to the upper-bounded complementarity system

$$0 \leq y \perp By + d + w \geq 0, \quad 0 \leq w \perp u - y \geq 0. \quad (7)$$

The base matrix B is a Z-matrix, since $A \geq 0$. The naive standard-LCP embedding of (7) has a $+I$ block and is not itself a Z-matrix LCP; the bounded form is the correct structural statement. We use this complementarity view to explain the sign-balanced structure, but our polynomial-time claim is restricted to the contraction/M-matrix regime of Proposition 2.

Remark 1. The large-impact bipartite regime is an interesting open case. The transform places it in the Tarski/CLS world and gives the upper-bounded Z-matrix structure above, but we do not claim a polynomial-time algorithm for arbitrary $\rho(A) \geq 1$. This does not contradict polynomial algorithms for monotone payment-network clearing [1]. Those algorithms exploit payment-network structure such as no-overpayment/no-fraud constraints and conservation-like flow or circulation properties. The transformed DeFi map is monotone and piecewise-linear, but it is a price-impact liquidation map rather than a payment allocation map; its feedback matrix need not be substochastic or conservative, and in the large-impact regime it may have spectral radius at least one. In the small-impact regime where this feedback becomes contractive, Proposition 2 recovers polynomial-time computation.

4 Membership and All-or-Nothing Clearing

Proposition 3. *For rational inputs and rational $\delta > 0$, smooth δ -clearing belongs to PPAD.*

Proof. The right-hand sides of the price and liquidation equations are represented by a rational piecewise-linear circuit over addition, multiplication by rational constants, and min/max gates. The price clipping and liquidation clipping make this circuit a continuous self-map of a compact rational box. A δ -clearing is precisely an approximate fixed point of this piecewise-linear map, up to a harmless rescaling of residuals between price and liquidation coordinates. Approximate fixed points of rational piecewise-linear FIXP circuits are in PPAD via the Linear-FIXP/PPAD correspondence [9].

Proposition 4. *The all-or-nothing clearing existence problem belongs to NP. If the signed feedback structure is balanced, all-or-nothing clearing can be found in polynomial time.*

Proof. For NP membership, a certificate is the set of liquidated positions. Given this set, prices are computed directly from (3), and each threshold condition can be checked. In the balanced case, after the sign transform the induced map on $\{0, 1\}^m$ is monotone. Starting from the bottom element and iterating the map, the sequence is coordinate-wise nondecreasing in the transformed order. Since every coordinate is binary, each coordinate changes at most once, and the iteration stabilizes after at most m strict changes. This is the finite-lattice analogue of the monotone-clearing algorithms used in classical financial networks; unlike the smooth case, finiteness gives an immediate polynomial bound.

5 PPAD-Hardness in the Non-Bipartite Regime

Our hardness proof uses only two generalized-circuit gates. Let $T_{[0,1]}(t) = \min\{\max\{t, 0\}, 1\}$. A κ -approximate assignment to such a circuit assigns a value in $[0, 1]$ to every wire and makes the output of each gate differ from its prescribed value by at most κ . This is a local notion of satisfaction: circuits may contain cycles, and no topological evaluation order is assumed.

Lemma 3 (SUM/NOT source). *There is a constant $\kappa > 0$ such that finding a κ -approximate solution to generalized circuits with only the gates*

$$G_+(u, w) : y_v = T_{[0,1]}(y_u + y_w), \quad G_{1-}(u) : y_v = 1 - y_u$$

is PPAD-complete.

The lemma is Proposition 5.5 in the journal version of Filos-Ratsikas et al. [5]. Budish et al. [6] give a self-contained SUM/NOT simplified-circuit theorem with fan-out two. This restricted source is important: because truncated-SUM is already a primitive, our reduction does not build high-gain chains, and the error amplification below is independent of circuit size. In particular, the reduction uses neither comparison gates nor a band-reset construction; all gadgets below have constant self-feedback bounded away from one.

Lemma 4 (unbalanced SUM/NOT source). *Lemma 3 remains PPAD-hard under the promise that the signed dependency graph of the source circuit contains a cycle with an odd number of NOT gates in the same connected component as an original hard-instance wire.*

Proof. Given a SUM/NOT circuit C , pick any wire s in it. Add two fresh wires a, b and the two gates

$$a = 1 - s, \quad b = T_{[0,1]}(s + a).$$

Every solution of the augmented circuit restricts to a solution of C , because the original gates are unchanged. Conversely, any exact solution of C extends by setting $a = 1 - s$ and $b = 1$. The new dependency graph contains the cycle $s - a - b - s$, with the edge $s - a$ coming from a NOT gate and the two edges incident to b coming from the SUM gate. Thus the cycle has odd NOT parity and lies in the connected component containing s . A κ -approximate solution of the augmented circuit gives a κ -approximate solution of C by projection, so a solver for this promised subclass would solve the PPAD-hard source problem.

Wire encoding and writer equation. For each circuit wire v , create an asset and encode $y_v = 2 - p_v$. Intended prices lie in $[1, 2]$, inside the global clipping box $[0, 3]$. A collateral-only writer with collateral output O , debt input U , and parameters C, D, ε satisfies

$$x = \Pi_{[0,1]}(c - ay_U + \lambda y_O), \quad a = D/\varepsilon, \quad \lambda = C/\varepsilon, \quad c = (2D - 2C)/\varepsilon. \quad (8)$$

If this writer is the unique effective writer of O , then $y_O = x$. When $\lambda < 1$, the gadget defines a unique functional output and approximate-clearing residuals are amplified by at most a constant factor. All gadgets below have $\lambda \leq 1/2$ or $1/3$, except for no high-gain construction at all. The wire band is invariant: unique writers output $x \in [0, 1]$, and the only co-writing operation is the self-bounding average gadget.

The construction respects a close-factor bound. If a writer should contribute η units of output price drop, take $C = 1$, $\alpha = \varphi/2$, and output slope $s_O = 2\eta/\varphi$. Then $s_O\alpha = \eta$ and $\alpha \leq \varphi C$. Thus coefficients are implemented through market depth, not by selling more collateral than a position contains.

The two gadgets. The NOT gate $y_O = 1 - y_U$ uses one writer with $C = 1, D = 2, \varepsilon = 3$, giving

$$y_O = \Pi_{[0,1]} \left(\frac{2 - 2y_U + y_O}{3} \right),$$

whose unique fixed point is $1 - y_U$. The truncated-SUM gate is a constant-size macro. First, two co-writers, each contributing half a unit of price drop and reading the complement wires $\bar{U}, \bar{W}$, create an average asset A satisfying $y_A = (y_U + y_W)/2$. Second, a NOT gadget creates $\bar{A}$. Finally, one writer with $C = 1, D = 2, \varepsilon = 2$ reads $\bar{A}$ and satisfies

$$y_O = \Pi_{[0,1]} \left(y_A + \frac{1}{2} y_O \right),$$

whose unique fixed point is $T_{[0,1]}(2y_A) = T_{[0,1]}(y_U + y_W)$. The algebra and constant error bounds are recorded in Appendix B. In particular, a source gate violation is at most $B\delta$ for an absolute constant B independent of the circuit size.

Lemma 5 (forward decoding). *For every SUM/NOT circuit C , one can construct in polynomial time a collateral-only liquidation instance I_C such that every δ -clearing of I_C decodes via $y_v = 2 - p_v$ to a $B\delta$ -satisfying assignment of C , where $B = O(1)$ is independent of $|C|$.*

Proof. Replace each source gate by the corresponding constant-size macro. Debt-leg fan-out is non-invasive: an asset can be read as debt by many downstream writers without changing its price, since only collateral sales enter (3). Every public wire has a unique effective writer, except for the two intended co-writers inside the average macro. All self coefficients are bounded away from one, and every macro has constant size, so the error amplification is a universal constant. Generalized-circuit satisfaction is local over gates, hence no topological induction is needed even when the source circuit contains cycles.

The target search problem is total: clipping makes the clearing map continuous on a compact box, so Brouwer gives at least one exact clearing. A total-search reduction therefore needs only the forward decoding property above; we do not prove that every approximate circuit solution extends to a clearing.

Lemma 6 (gadget parity). *Let H_C be the signed undirected dependency graph of a SUM/NOT circuit C , with a negative edge for every NOT gate and positive edges from each input of a SUM gate to its output. For the liquidation instance I_C constructed above, with no additional disconnected anchor, the asset borrowing graph $G_A(I_C)$ is bipartite if and only if H_C is sign-balanced. In particular, an odd-NOT cycle in C maps to an odd cycle in $G_A(I_C)$.*

Proof. The claim is a parity accounting for the gadget templates. A NOT gate $U \rightarrow O$ is implemented by one writer whose collateral is O and debt is U , hence it contributes an odd path, in fact a single collateral-debt edge, between the two terminal assets. A SUM incidence $U \rightarrow O$ is implemented through the path

$$U - \bar{U} - A - \bar{A} - O,$$

where $U - \bar{U}$ and $A - \bar{A}$ are NOT-gadget edges and the other two edges are writer edges inside the averaging and final-output gadgets. This path has length four, hence even. The same holds for the other input $W \rightarrow O$. Internal paths inside a SUM macro between its terminals are even, and the internal subgraph has no odd cycle.

Thus the borrowing graph is obtained from the signed dependency graph by replacing negative edges by odd-length paths and positive edges by even-length paths, together with bipartite internal subdivisions. By Harary's parity criterion, such a graph is bipartite exactly when every cycle in H_C has an even number of negative edges. Therefore sign-balance of the source circuit is equivalent to

bipartiteness of the constructed asset borrowing graph. The odd-NOT cycle supplied by Lemma 4 consequently gives an intrinsic non-bipartite cycle in the hard liquidation instance.

Constant approximation budget. Let B_- and B_+ be the constant error amplifications of the NOT and SUM macros, and let B dominate both. These constants do not depend on the number of circuit gates. This independence is the main payoff from reducing from the restricted SUM/NOT source problem: the DeFi construction replaces each source gate by a constant-size macro and never simulates a comparator by a long amplification chain. If the clearing residual is at most δ , then each decoded source gate is violated by at most $B\delta$. Taking $\delta_0 = \kappa/(10B)$, where κ is the source approximation constant, converts any δ_0 -clearing into a valid κ -solution of the source circuit.

Theorem 2. *There exists a constant $\delta_0 > 0$ such that, in the non-bipartite regime, finding a δ_0 -approximate clearing in collateral-only linear price-impact liquidation markets is PPAD-hard. Consequently, for a sufficiently small fixed δ_0 , the problem is PPAD-complete.*

Proof. Let κ be the constant from Lemma 4 and let B be the constant from Lemma 5. Choose $\delta_0 = \kappa/(10B)$. Given a promised unbalanced SUM/NOT circuit C , build I_C with no disconnected anchor. By Lemma 6, the asset borrowing graph of I_C is non-bipartite. Any δ_0 -clearing decodes to a κ -satisfying assignment of C . Thus a clearing oracle for non-bipartite instances would solve a PPAD-complete source problem. Membership follows from the PPAD proposition.

The quantifiers in Theorem 2 are worst-case. We do not claim that every non-bipartite borrowing graph makes clearing hard. Rather, Lemma 6 shows that the constructed hard instances are non-bipartite for an intrinsic reason: the source circuit contains odd-NOT frustration, and the liquidation gadgets preserve that parity as an odd collateral-debt cycle. The non-bipartiteness is therefore part of the hard component, not a disconnected certificate attached after the fact.

6 Related Work and Discussion

Financial clearing as Tarski fixed points. Eisenberg and Noe [14] gave the classical monotone clearing model, and Rogers and Veraart [15] studied default costs. Recent work has made the Tarski viewpoint explicit. Besting, Hoefer, and Huth [1] compute extremal Tarski fixed points in generalized Eisenberg–Noe networks with monotone piecewise-linear payment functions and default costs, obtaining a strongly polynomial algorithm for minimal clearing and a polynomial algorithm for maximal clearing. This line is the right foil for our positive result. Their model remains a payment-network model with no-overpayment/no-fraud style constraints and conservation-like structure. Our transformed bipartite map is monotone and piecewise-linear, but it is generated by price impact and health-factor liquidation. It can have super-unit feedback gain and need not satisfy

the substochastic or circulation structure used by payment-network algorithms. Thus the unconditional conclusion on the sign-balanced side is CLS/Tarski membership; polynomial-time computation follows under the separate small-impact condition.

Fire sales and lattice equilibria. Fire-sale models also connect price impact with lattice structure. Bertschinger et al. [2] study strategic fire-sale games and show that, for a broad range of price-impact functions such as convex impact, equilibria exist and form a complete lattice; they also analyze best-response convergence and cycling. Their setting is a strategic portfolio liquidation game. Ours is a DeFi lending clearing problem with collateral/debt roles, health-factor thresholds, and a graph-theoretic sign condition. We use Tarski structure only on the sign-balanced side, and we show that odd collateral-debt cycles can implement PPAD-hard SUM/NOT fixed-point structure.

CDS hardness and DeFi lending. Schuldenzucker, Seuken, and Battiston [16] showed PPAD-completeness of financial-network clearing with credit default swaps; Dohn, Hansen, and Klinkby [4] sharpened this hardness line and studied additional decision problems. These results show that derivative-based clearing has a rich PPAD/FIXP hardness theory. Our hardness uses no CDS, no default-contingent derivative payments, and no debt-side buy pressure. The NOT primitive is produced by collateral-only liquidation lowering the price of another position’s debt asset. On the DeFi side, empirical and theoretical work documents liquidations, price impact, and systemic fragility [10–13]; Bartoletti and Lipparini [3] give a formal model of DeFi lending protocols and their incentive mechanisms. We complement this literature by isolating the fixed-point clearing problem created by price-impact liquidation feedback.

DeFi interpretation and limitations. The model is a linear price-impact abstraction for aggregate liquidation pressure; constant-product AMMs and multi-pool routing are nonlinear and path-dependent. Extending the exact CLS/Tarski formulation or the gadget proof to those AMM models is open. The hardness theorem is worst-case: small-impact non-bipartite instances may still be contractions and easy.

Open complexity of the balanced large-impact case. The sign-balanced map is subtraction-free, monotone, and clipped-affine: it aggregates nonnegative weighted inputs and clips against constants, but lacks a primitive for comparing two variables. Known hard Tarski constructions typically use min/max or comparison primitives, which appear to require subtraction or anti-monotone dependence. At the same time, payment-network algorithms for monotone piecewise-linear clearing rely on conservation-like structure absent from large-impact price-impact liquidation. Thus we leave the polynomial-time status, or finer complexity classification, of large-impact sign-balanced clearing as an open problem.

Conclusion. Collateral-only DeFi liquidation creates a native source of nonmonotone clearing feedback. Bipartite sign-balance gives a monotone Tarski instance in CLS and a polynomial-time contraction problem under small impact; removing sign-balance enables SUM/NOT circuits and yields constant-approximation PPAD-hardness.

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A Additional Details for Sign-Balanced Clearing

This appendix records the complementarity conversion used as a structural observation in Section 3. For $y = \Pi_{[0,u]}(Ay + c)$, define $B = I - A$ and $d = -c$. Then $F(y) = By + d$. Coordinate-wise, the fixed-point condition is equivalent to

$$y_i = 0 \Rightarrow F_i(y) \geq 0, \quad 0 < y_i < u_i \Rightarrow F_i(y) = 0, \quad y_i = u_i \Rightarrow F_i(y) \leq 0.$$

Introducing an upper-bound multiplier $w \geq 0$ gives the upper-bounded LCP formulation

$$0 \leq y \perp By + d + w \geq 0, \quad 0 \leq w \perp u - y \geq 0.$$

Indeed, if $0 < y_i < u_i$, then $w_i = 0$ and $(By + d)_i = 0$. If $y_i = 0$ and $u_i > 0$, then $w_i = 0$ and $(By + d)_i \geq 0$. If $y_i = u_i$, then $(By + d)_i = -w_i \leq 0$. Coordinates with $u_i = 0$ may be eliminated in preprocessing.

The ordinary standard-LCP embedding of this upper-bounded system, with variables (y, w) , has matrix

$$\begin{pmatrix} B & I \\ -I & 0 \end{pmatrix}.$$

The $+I$ block is forced by the sign of the upper-bound multiplier and gives positive off-diagonal entries. Hence the bounded system should not be described as an ordinary Z-matrix LCP. What bipartiteness gives unconditionally is the monotone clipped-affine form, the Tarski reduction, and the upper-bounded Z-matrix base structure. A polynomial-time algorithm follows in the small-impact regime where $\rho(A) < 1$, equivalently $I - A$ is a nonsingular M-matrix.

B Gadget Algebra and Error Bounds

We use the encoding $y_v = 2 - p_v$. For a writer with collateral O , debt U , and parameters C, D, ε , the equation is

$$x = \Pi \left(\frac{2D - 2C}{\varepsilon} - \frac{D}{\varepsilon} y_U + \frac{C}{\varepsilon} y_O \right).$$

If it is the unique writer of O , then $y_O = x$. If $C/\varepsilon < 1$, the right-hand side is a contraction in y_O , and any residual δ gives an output error at most $\delta/(1 - C/\varepsilon)$ up to the harmless constant accounting for price and liquidation residuals.

For the NOT gadget, $C = 1, D = 2, \varepsilon = 3$. Hence

$$y_O = \Pi \left(\frac{2 - 2y_U + y_O}{3} \right),$$

whose unique fixed point is $1 - y_U$. The self coefficient is $1/3$, so a conservative error bound is $B_- = 3$.

For the average gadget, first maintain complement wires $\bar{U}, \bar{W}$ by NOT gadgets. Create an average asset A with two co-writers, both with collateral A , each

contributing $\eta = 1/2$ units of output price drop. Writer 1 reads $\bar{U}$, writer 2 reads $\bar{W}$, and both use $C = 1, D = 2, \varepsilon = 3$. Since $y_{\bar{U}} = 1 - y_U$, writer 1 satisfies

$$x_1 = \frac{2}{3}y_U + \frac{1}{3}y_A,$$

and similarly $x_2 = \frac{2}{3}y_W + \frac{1}{3}y_A$. The output equation is $y_A = (x_1 + x_2)/2$, so $y_A = (y_U + y_W)/2$. The self coefficient is again bounded away from one, and the error is a constant multiple of δ .

The final gain-2 clipped writer reads $\bar{A}$ and uses $C = 1, D = 2, \varepsilon = 2$. Since $p_{\bar{A}} = 2 - (1 - y_A) = 1 + y_A$, the writer equation becomes

$$y_O = \Pi \left(y_A + \frac{1}{2}y_O \right).$$

If $y_A \leq 1/2$, the unique fixed point is $2y_A$; if $y_A > 1/2$, the unique fixed point is 1. Thus the output is $T_{[0,1]}(2y_A) = T_{[0,1]}(y_U + y_W)$. All self coefficients are at most $1/2$ or $1/3$ inside the macro, so the total error bound B_+ is an absolute constant.

The close-factor bound can be met by scaling the output impact. For a writer with desired output contribution η , take collateral amount parameter $C = 1$, liquidation amount $\alpha = cf/2$, and slope $s_O = 2\eta/cf$. Then $s_O\alpha = \eta$ and $\alpha \leq cf \cdot C$.

C Source Problem and Fan-Out

The restricted source problem in Lemma 3 is taken as a black box. The SIAM Journal version of Filos-Ratsikas et al. [5] states PPAD-completeness for constant-approximate generalized circuits with gate set $\{G_+, G_{1-}\}$. Budish et al. [6] give an alternative self-contained reduction to simplified circuits containing only SUM and NOT gates, even with bounded fan-out. Our construction does not require bounded fan-out because debt-leg reads are non-invasive.

The unbalanced-source promise used in Lemma 4 is not a target-side graph tag. It is obtained before translation by adding the two gates $a = 1 - s$ and $b = T_{[0,1]}(s + a)$ through an original source wire s . This creates a signed source cycle $s - a - b - s$ with one NOT edge. Lemma 6 then maps that source frustration to an odd collateral-debt cycle in the constructed borrowing graph.

D Subtraction-Free Structure of the Balanced Side

The sign-balanced map in Theorem 1 has the special form $y = \Pi(Ay + c)$ with $A \geq 0$. It is therefore monotone and subtraction-free: it can form nonnegative weighted sums and clip against constants, but it has no direct primitive for comparing two variables. This explains why we state a CLS upper bound rather than a matching Tarski-hardness theorem. Boolean monotone fragments such

as AND/OR have easy least fixed points by finite Kleene iteration, while large-domain Tarski hardness constructions typically use min/max or other variable-comparison operations. Implementing such operations appears to require an anti-monotone dependence, which is precisely what the bipartite sign transform rules out. We leave the exact complexity of the large-impact sign-balanced subclass open.